

INDIAN STOCK MARKET EDUCATION PLATFORM

Stocktute

Full project guide — what has been built, how it is wired,
and what remains.

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Educational content only. Not investment advice.
Past patterns do not guarantee future results. Virtual money only.

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1. Product in one page

Stocktute is an Indian-market education platform for NSE/BSE. Students learn charts, candlesticks, risk, and process with **virtual money only**. The product never gives real buy/sell advice, stock tips, targets, or profit promises. A 12-level curriculum controls vocabulary and which tools unlock. Futures and options stay locked until the student passes Level 6 risk training.

Two connected teaching features sit on top of a deterministic detection engine:

- **AI Classroom** — one historical chart taught in six layers (What, Why, Context, Predict, Reveal, Failure).
- **Everyday Market Chapter** — a draft chapter generated after the cash-market close from that session's engine facts, then reviewed by an admin before it can go live.

The most important product rule is technical: **the language model must never detect patterns or read prices**. Code computes ChartFacts JSON. The tutor only explains those facts. If a number is missing, the copy says “I don't have that data.”

2. Non-negotiable rules

- Never promise or imply guaranteed profits.
- A candlestick pattern is a label, not a buy or sell signal.
- Every strategy study must include invalidation and a position-sizing rule.
- F&O stays locked until Level 6 risk training is passed.
- Paper trading comes before any live broker integration.
- Every generated chapter carries the disclaimer: Educational content only. Not investment advice. Past patterns do not guarantee future results.
- Psychology words such as “buyers” and “sellers” are allowed. Standalone “buy”, “sell”, “target”, “sure-shot”, “guaranteed”, and “will go up” are rewritten by the compliance filter.

3. Tech stack

Layer	Choice	Notes
App	Next.js 15.5 App Router + React 19 + TypeScript	src/app
Auth	NextAuth v5 (JWT) — email OTP + Google	src/lib/auth, src/auth.ts
Database	MongoDB via Mongoose 9	src/lib/db/mongoose.ts
OTP / cache	Upstash Redis	10-minute OTP TTL
Charts	TradingView Lightweight Charts v5	src/components/charts
Detection	In-house TypeScript engine	src/lib/detection
Jobs	tsx workers	price feed + daily chapter
Tests	Vitest	44 tests passing as of 22 Sep 2026
Styling	Tailwind CSS 4 + design tokens	src/styles/tokens.ts
Payments (schema only)	Razorpay keys in env	Billing page is a placeholder
LLM (not wired yet)	Anthropic SDK in package.json	Classroom uses a fact-grounded template narrator

Table 1. Stack as implemented in the repository.

4. Core architecture

All teaching numbers flow one way. The UI and the tutor are not allowed to invent OHLC, pattern names, EMA, RSI, support, or outcomes.

```
OHLCV bars
→ detection engine  analyzeChart()
→ ChartFacts JSON   (setups, context, forward returns, base rates, missingFacts)
→ narrator / daily copy  (templates; LLM later may only explain this JSON)
→ compliance filter
→ MongoDB
→ UI (classroom, journal, admin review, later student daily chapter)
```

Figure 1. Teaching pipeline. Code detects. Copy explains.

If a value cannot be computed it is **null** and listed in missingFacts. Confirmation is the next **closed** bar's close beyond the pattern high/low. If that bar does not exist, confirmation is unavailable. Base rates are never invented: sampleSize 0 and hitRate null when no universe is supplied.

5. Build status (steps 1–6)

Step	Deliverable	Status
1	Detection engine with unit tests (12 patterns, EMA, RSI, volume, S/R, outcomes, base rates)	Done
2	AI Classroom 6-layer flow on one historical RELIANCE daily hammer	Done
3	Persist predictions + mistake journal with Play/Pause/Step replay	Done
4	Daily generation job, compliance filter, admin review (draft / approve / reject / publish)	Done
5	Student Everyday Market view: level gating, streaks, archive	Not built
6	Explain-back grading, misconception detector, spaced repetition, calibration chart	Not built

Table 2. Ordered delivery plan from the original product spec.

Older platform shells still exist as placeholders: Simulator, Replay, Leaderboard, Billing. Curriculum, skill tree, lessons, drills, and the risk calculator are live.

6. Full repository tree

Only source that matters for the current product is shown. node_modules and .next are omitted.

```
stocktute/
package.json           next, vitest, workers
.env.example
docs/
  phase1-architecture.md  early shell notes (partly superseded)
  Stocktute-Project-Guide.pdf  this document
src/
  auth.ts
  app/
    layout.tsx, globals.css
    (marketing)/page.tsx
    (auth)/login verify
    (app)/
      dashboard skill-tree  requires session
      classroom lessons/[slug]
      classroom/live 6-layer tutor
      journal sandbox tape
      drills risk formulas mistake journal
      admin chapter list + generate
      admin/chapters/[date] review / edit / approve
      simulator replay leaderboard billing placeholders
  api/
    auth/[...nextauth] auth/otp/request
```

```

    candles/[symbol] stream health
    predictions
    admin/chapters/generate
    admin/chapters/[date]
    admin/settings
components/
  admin/      AdminGenerate ChapterReview
  charts/     LightweightChart
  learn/      ClassroomTutor PatternReplay JournalClient
              SkillTreeView PatternWorkbookView PositionSizeCalculator
  layout/     Sidebar TickerRail ComplianceBanner
lib/
  detection/  engine patterns indicators trend levels
              outcomes baseRates classroomLesson narrate
  journal/    score teachingCharts
  daily/      calendar tapes select copy compliance
              generate persist universe positionSize
  curriculum/ catalog content
  priceFeed/  MockNseProvider candleBuilder generateHistory
  auth/       otp options requireAdmin
  db/mongoose.ts
models/
  User Prediction MistakeJournalEntry
  DailyChapter ComplianceEvent AppSettings
  Candle Lesson Question Order Position Trade
workers/
  priceFeedWorker.ts dailyChapterWorker.ts
tests/
  detection/  engine.test.ts classroomLesson.test.ts
  journal/    score.test.ts
  daily/      generate.test.ts
  curriculum/ catalog.test.ts
  detector/   candlestickPatterns.test.ts (older live-tape labels)

```

7. Pages and routes

Path	Audience	What it does
/	Public	Marketing landing
/login, /verify	Public	Email OTP sign-in
/dashboard	Student	Welcome, 12-level path, paper wallet figure
/skill-tree	Student	Levels 1–12 with lock reasons
/lessons, /lessons/[slug]	Student	150+ lesson pages; candlestick academy workbooks
/classroom	Student	AI Classroom — 6-layer historical hammer lesson
/classroom/live	Student	Sandbox live tape with pattern markers
/journal	Student	Wrong 5-bar guesses + chart replay + habit notes
/drills	Student	Short quizzes
/risk	Student	Position-size calculator ($\text{capital} \times \text{risk\%} \div \text{distance}$)
/formulas	Student	Formula-builder preview
/admin	Admin only	Generate daily drafts, auto-publish toggle, chapter list
/admin/chapters/[date]	Admin only	Edit copy, charts, quiz, approve / reject / publish
/simulator /replay /leaderboard /billing	Student	Phase placeholders — not implemented

Table 3. App Router pages. (app) layout redirects unsigned users to /login.

Admin pages call `requireAdmin()`. Students with role “student” are sent to `/dashboard`. The sidebar shows an Admin link only when `session.user.role` is admin.

8. API surface

Method	Path	Auth	Purpose
POST	/api/auth/otp/request	Public	Email a 6-digit code (Redis TTL 10 minutes, IST timestamps)
*	/api/auth/[...nextauth]	Public	NextAuth Google + email-OTP credentials
GET	/api/candles/[symbol]	Session via app	History; falls back to generated OHLC if Mongo is thin
GET	/api/stream	Session	SSE live quotes
GET	/api/health	Public	Health check
POST	/api/predictions	Signed-in	Lock classroom prediction; score from engine; journal misses
POST	/api/admin/chapters/generate	Admin or CRON_SECRET	Build and upsert DailyChapter draft
PATCH	/api/admin/chapters/[date]	Admin	Save edits (re-filter), approve, reject, publish
POST	/api/admin/settings	Admin	autoPublishDaily toggle

Table 4. HTTP APIs.

Classroom predictions do not trust client prices. The client sends only chartKey, direction, confidence (50–100), and a reason. The server resolves the teaching chart, runs scorePrediction against ChartFacts, and upserts one Prediction per user per chart. A miss writes MistakeJournalEntry; a later hit on the same chart deletes that journal row.

9. Detection engine (step 1)

Entry point: analyzeChart({ candles, symbol, timeframe, baseRates? }) in src/lib/detection/engine.ts. It returns ChartFacts: last close, EMA 20/50, RSI(14), volume average, support/resistance arrays, setups[], and missingFacts[].

Each DetectedSetup includes

- Pattern name, bias (bullish / bearish / neutral), expectedMove (up / down / sideways).
- Inclusive startIndex and endIndex on the candle array.
- Key levels: open, high, low, close, invalidation.
- Confirmation: confirmed, unconfirmed, or unavailable (no next closed bar).
- Context: prior 20-bar trend, volume vs 20-bar average, EMA 20/50, RSI(14), nearest S/R and distances.
- Forward close returns at 1, 3, 5, 10 bars (null if those bars are not closed).
- Base rate: universe, sampleSize, hitRate (null when sampleSize is 0).

Twelve textbook patterns

Doji, Hammer, Hanging Man, Inverted Hammer, Shooting Star, Bullish Engulfing, Bearish Engulfing, Piercing Line, Dark Cloud Cover, Morning Star, Evening Star, Three White Soldiers, Three Black Crows. Geometry lives in geometry.ts; scanners in patterns.ts. One label per end bar, ranked so multi-bar patterns outrank single-bar wicks when both fire.

Tests: tests/detection/engine.test.ts (23) plus classroomLesson.test.ts. First-run issues that were fixed: oxc parse on defaulted destructuring, Three White Soldiers outranking hammers (now require body/range > 0.45), and

hanging-man vs hammer depending on the prior bar's direction.

10. AI Classroom (step 2)

Route /classroom is a server page that builds buildHammerClassroomLesson(studentLevel). Level is read from User.level (default 2) and only changes stop-loss wording. The live tape was moved to /classroom/live so the graded lesson is always historical.

Layer	Student action	Engine constraint
1 What	See highlighted hammer candles and OHLC in rupees	Name and prices from DetectedSetup
2 Why	Read rejection story on that bar only	Low / close / high / volume vs average from facts
3 Context	Compare strong vs weak twin side by side	Same pattern name, different neighbourhood
4 Predict	Must pick up / down / sideways, 50–100% confidence, ≥8 character reason	Future bars hidden; cannot skip
5 Reveal	Play / Pause / Step the next bars	5-bar close % and base rate; one chart is an anecdote
6 Failure	See a hammer that did not follow through	Invalidation print; stop-loss glossed if level < 6

Table 5. Six-layer tutor. Narration is src/lib/detection/narrate.ts (templates, not Anthropic).

Highlighting is a lime border on pattern bars (highlightRange on LightweightChart). The teaching chart key is classroom:hammer:RELIANCE:1D:primary. If a fixture is not actually a Hammer, the builder throws — it will not invent a detection.

11. Predictions and mistake journal (step 3)

Locking a prediction POSTs to /api/predictions. Scoring is src/lib/journal/score.ts. “Wrong” means the student’s direction does not match the engine’s 5-bar close bucket (up if return > 0.5%, down if < −0.5%, else sideways). A missing 5-bar close is not a miss.

Mistake tags (only on misses)

- wrong-direction — bucket mismatch.
- followed-textbook-bias — guessed the pattern’s expectedMove.
- ignored-volume — textbook side while volume vs average < 1.
- ignored-context — reason did not mention trend, volume, support, EMA, RSI, wick, or invalidation.
- certainty-language — always / definitely / sure-shot / guaranteed / will go.

/journal lists misses with the stored candle snapshot. PatternReplay is the same Play / Pause / Step control as Reveal. Habit notes appear only when the same tag shows at least twice. Correct guesses stay off this list on purpose.

12. Everyday Market backend (step 4)

After a cash-market session the job generateDailyChapter(date) runs. Weekends and a stub NSE holiday list are skipped. There is no licensed EOD vendor yet: indices use generateHistory; a few constituents get fixture overlays so the picker always has a clean pattern and a trap without inventing detections.

Section	Min level	Source of numbers
Market Story	1	NIFTY and BANKNIFTY last bar OHLC, change vs prior close, volume vs 20-bar average. No news invented.

Section	Min level	Source of numbers
Pattern of the Day	2	Highest-quality recent setup from the engine.
Yesterday's Predict-and-Reveal	2	Pattern ending before the last closed bar; hideAfterPattern true.
Spot It Yourself	3	Three mixed-bias snapshots; labels stored for later student UI.
Trap of the Day	4	Same drawing, weak context (light volume or far from S/R).
Risk Drill	6	Virtual $1,00,000 \times 1\% \div [\text{close} - \text{invalidation}]$.
Rule Check	7	Last close above EMA(20) and volume > 1.5× average. Educational list only.
Daily Quiz	1	Five items from the pattern-of-day facts (MCQ, chart-mark, explain).

Table 6. Daily chapter sections. Student gating is step 5 — today only admins see the draft.

Compliance filter

src/lib/daily/compliance.ts rewrites phrases before save and again when an admin edits. Hits are stored on the chapter and in ComplianceEvent. Auto-publish (AppSettings.autoPublishDaily) only publishes when blockedCount is 0; otherwise the chapter stays draft.

Phrase	Becomes
buy / sell (word boundary; not buyers/sellers)	study
target(s)	reference level
sure-shot	study case
guaranteed	not promised
will go up / will go down	moved in the textbook direction in past samples
multibagger, book profit, accumulate	study example / review the invalidation / watch

Table 7. Compliance rewrites.

How to generate

- Admin UI: /admin → Generate draft (optional session date).
- CLI: npm run daily-chapter (optional YYYY-MM-DD argument).
- Cron: POST /api/admin/chapters/generate with Authorization: Bearer \$CRON_SECRET. Intended ~18:00 IST.

Universe stand-in: NIFTY, BANKNIFTY, RELIANCE, TCS, HDFCBANK, INFY, ICICIBANK, HINDUNILVR, SBIN, BHARTIARTL, ITC, KOTAKBANK.

13. Curriculum (12 levels)

Level	Name	Notes
1	Market basics	NSE/BSE, orders, accounts — open
2	Charts and candlesticks	Classroom default; candlestick academy workbooks
3	Patterns and indicators	Open lessons
4	Price action	Catalog present
5	Fundamentals	Catalog present

Level	Name	Notes
6	Risk and position sizing	Unlocks F&O later; calculator live
7	Rule-based strategies	Daily rule-check is tagged L7
8	Backtesting	Outline
9	Paper trading	Simulator page still placeholder
10	F&O	Locked until Level 6 risk test
11	Algo trading	Locked / later
12	Capstone + certificate	Later

Table 8. Catalog in `src/lib/curriculum`. 150+ unique slugs; F&O lessons stay locked.

14. MongoDB models

Collection	Role
User	Email, Google/email provider, tier free/lab, wallet ■5,00,000 virtual, level 1–12 (default 2), xp, language en/hi/hinglish, streak, role student/admin
Prediction	One locked predict per user per chartKey: direction, confidence, reason, actual 5-bar bucket, tags, engine snapshot
MistakeJournalEntry	Misses only, unique per user+chartKey, replay snapshot
DailyChapter	YYYY-MM-DD IST, status draft/approved/rejected/published, sections[], compliance log, reviewer
ComplianceEvent	Rewrite audit trail (phrase, original, rewritten)
AppSettings	Singleton key=site, autoPublishDaily
Candle	OHLCV by symbol+timeframe+time (1m/5m/15m/1D)
Lesson / Question	Older CMS-style schemas; current syllabus is mostly the TypeScript catalog
Order / Position / Trade	Paper-trading schema ready; simulator UI not built

Table 9. Mongoose models in `src/models`.

15. Auth, wallet, price feed

Auth

OTP is stored in Redis and consumed only after a successful Mongo upsert, so a brief Atlas blip does not burn a valid code. Credentials errors map to `invalid_otp` and `database_unavailable`. Email timestamps are formatted in Asia/Kolkata. Google sign-in upserts User with provider google.

Wallet

Every user starts with a ■5,00,000 virtual wallet. Nothing in the classroom or daily chapter places a live order. The risk drill uses a hypothetical ■1,00,000 × 1% example independent of the wallet.

Price feed

`npm run worker` polls `MockNseProvider`, builds 1m/5m/15m/1D candles, writes closed bars to Mongo, and publishes quotes to Redis for `/api/stream`. If stored history is too thin or all bars are single-tick (OHLC equal),

/api/candles falls back to generateHistory so charts have real bodies and wicks. That fallback is what used to look like a dotted line on 1-minute classroom charts.

16. How to run locally

```
npm install
# copy .env.example → .env and fill Mongo, NextAuth, Redis, SMTP
npm run dev           # http://localhost:3000
npm test              # vitest (44 tests)
npm run worker        # optional mock tape
npm run daily-chapter # optional: generate today's IST draft (needs Mongo + admin settings)
```

- Whitelist this machine's IP on Atlas if OTP sign-in fails with database_unavailable.
- Set a User.role to “admin” in Mongo to open /admin.
- Set CRON_SECRET to call the generate API from a scheduler at ~18:00 IST.
- ANTHROPIC_API_KEY is unused for teaching copy today; step 2–4 use templates grounded in ChartFacts.

17. Tests

File	What it locks
tests/detection/engine.test.ts	12 patterns, EMA/RSI, trend, S/R, forward returns, base rates
tests/detection/classroomLesson.test.ts	Hammer on strong/weak/failed tapes; narrator uses engine rupees; no live-trade wording
tests/journal/score.test.ts	Match vs miss, textbook-bias tags, habits after two similar misses, teaching-chart registry
tests/daily/generate.test.ts	Weekend/holiday skip, eight sections on 2026-09-22, no buy/sell, risk shares, compliance rewrites
tests/curriculum/catalog.test.ts	12 levels, unique slugs, F&O locked
tests/detector/candlestickPatterns.test.ts	Older sandbox pattern labels on the live tape

Table 10. Vitest suites. Last full run: 6 files, 44 passed.

18. Open assumptions and next steps

Assumptions still open

- No licensed NSE EOD vendor. Daily chapters use synthetic 80-day tapes plus fixture overlays.
- Ten names stand in for NIFTY 50 constituents. Base rates are tallied on that synthetic universe, not a 5-year official history.
- NSE holiday calendar is a stub list in src/lib/daily/calendar.ts — replace with the official circular.
- Confirmation = next closed close beyond pattern high/low.
- User.level is stored but Everyday Market gating is not in the student UI yet.
- English-only classroom copy; language field exists (en / hi / hinglish) but is unused in the tutor.

Step 5 (not built) — Student Everyday Market

- Student route (for example /market) showing only sections at or below User.level; higher sections preview as “Unlock at Level X”.
- Streaks and XP for completing the daily chapter.
- Archive searchable by pattern name and date.
- Predict-and-reveal on yesterday's chart using already-closed data only.

Step 6 (not built) — Mastery loop

- Explain-back: student restates the idea; rubric (correct idea, condition, mentions risk).
- Misconception detector (for example “hammer always means up”).
- Spaced repetition at 1, 3, 7, 21 days with fresh engine snapshots.
- Confidence calibration chart: are 80% guesses right about 80% of the time?
- Optional LLM explanations still bound to ChartFacts, then the same compliance filter.

End of guide. Stocktute remains a virtual classroom. Nothing in this document is a recommendation to trade a live stock.